prfaq Plugin: Architecture Overview

Generated from codebase analysis

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1. What It Is

`prfaq` is a Claude Code plugin (v1.7.2) that implements Amazon's Working Backwards PR/FAQ process. It generates, reviews, stress-tests, and iterates on product discovery documents inside the terminal. The output is a professionally typeset LaTeX PDF (or Word `.docx` via pandoc).

The plugin contains no runtime code. It is composed entirely of prompt engineering artifacts—skill definitions, agent system prompts, reference guides, LaTeX templates, and shell scripts. Claude Code is the runtime; the plugin provides domain expertise and workflow structure.

2. Project Structure

Everything the plugin ships lives under `plugin/`, which is the plugin root. The layout serves Claude Code's `git-subdir` marketplace source (`"path": "plugin"`): once the entry is re-pointed, an install fetches that subtree and nothing else. `${CLAUDE_PLUGIN_ROOT}` resolves to it, and the paths below are relative to it. Six component categories:

`.claude-plugin/plugin.json`
: Plugin manifest declaring name, version, and description.

`skills/prfaq/`
: One skill (`SKILL.md`) orchestrating the six-phase generation workflow, plus ten reference guides encoding domain knowledge.

`agents/`
: Eight specialized subagents, each with a narrow role and specific reference guide dependencies.

`commands/`
: Fourteen slash commands for post-generation operations (review, meeting, feedback, export, vote, etc.).

`assets/`
: Two LaTeX templates: one for PR/FAQ documents, one for externalized press releases. Also includes a styled Word reference document for `.docx` export.

`scripts/`
: Five shell scripts handling PDF compilation, Word export, LaTeX preprocessing for pandoc, reference document generation, and project permission grants.

The repo root holds development material a consumer never receives—the dogfood `prfaq.tex`, `docs/`, `research/`, `meetings/`, `tests/`, and this repo's own release scripts in a second, unshipped `scripts/`.

3. The Skill

The single skill (`/prfaq`) is a 280-line prompt that orchestrates document generation across six phases:

1. **Phase 0: Research Discovery** — Scans `./research/` for primary data, invokes the researcher agent, builds a `.bib` file.
2. **Phase 1: Discovery** — Interactive questioning: stage, customer, problem, solution, differentiation, market, risks.
3. **Phase 2: Draft PR** — Generates the eight press release sections from the LaTeX template and `pr-structure.md`.
4. **Phase 3: Draft FAQ** — External FAQs, internal FAQs (by risk category), four-risks assessment, feature appendix, then peer review.
5. **Phase 4: Compile** — Runs `compile_prfaq.sh`, fixes overfull hbox warnings, iterates until clean.
6. **Phase 5: Review** — Evaluates the document against seven review criteria and identifies the weakest sections.

The skill also handles revise mode: if `prfaq.tex` already exists, it offers surgical edits instead of regenerating from scratch.

4. The 13 Commands

Each command is a focused, single-purpose operation that typically invokes one agent and handles its output:

Command	Purpose	Agent Used
`/prfaq:import`	Parse existing doc, launch workflow	researcher
`/prfaq:badge`	Stage-colored README badge	(none)
`/prfaq:review`	Standalone peer review	peer-reviewer
`/prfaq:meeting`	Interactive 4-persona review	4 meeting agents
`/prfaq:meeting-hive`	Autonomous consensus meeting	4 meeting agents
`/prfaq:meeting-listen`	Voiced playback of meeting	(none, uses TTS)
`/prfaq:feedback`	Surgical redraft from directives	feedback
`/prfaq:research`	Evidence search	researcher
`/prfaq:streamline`	Remove redundancy and weasel words	streamliner
`/prfaq:vote`	Three-gate go/no-go decision	(inline)
`/prfaq:export`	Convert to `.docx` via pandoc	(none)
`/prfaq:externalize`	Customer-facing press release	(inline)
`/prfaq:feedback-to-us`	Anonymous satisfaction rating	(none)

The intended workflow chains these sequentially: generate → badge → review → meeting → feedback (iterate) → streamline → vote → export → externalize.

Agent	Role	Key Guides Loaded
peer-reviewer	Adversarial review (Kahneman framework)	All 10 guides
researcher	Evidence search (local, quarry, web)	None
feedback	Cascading redraft engine	Context-dependent
streamliner	Scalpel editor for prose tightening	precise-writing
meeting-engineer	Wei: feasibility risk	principal-engineer, four-risks
meeting-customer	Priya: value risk, customer reality	ux-bar-raiser, four-risks
meeting-executive	Alex: strategic fit, devil's advocate	unit-economics, four-risks
meeting-builder	Dana: ambition risk, cost of inaction	four-risks, common-mistakes

5. The Eight Agents

The four meeting personas embody competing priorities. Wei and Alex pull toward caution; Dana pulls toward ambition; Priya grounds both sides in customer reality. In interactive mode (`/prfaq:meeting`), the user resolves disagreements. In hive mode (`/prfaq:meeting-hive`), a door-weighted consensus mechanism resolves them autonomously.

6. The Ten Reference Guides

Reference guides are the shared knowledge layer. Each is a standalone document encoding a framework, evaluation criteria, and stage-calibrated expectations. Agents load only the guides they need.

Every guide follows the same structure: framework definition, concrete signals and criteria, what to check in the PR/FAQ, and a stage calibration table that adjusts expectations for hypothesis, validated, and growth stages.

6.1 Document Structure Guides

`pr-structure.md`
: Section-by-section blueprint for the mock press release. Defines the eight sections (headline through call to action) with strong/weak examples and common failures. Establishes that the customer quote is fictional at all stages by design.

`faq-structure.md`
: Organization of external FAQs (5 required customer-facing questions) and internal FAQs (organized by value/market, technical, and business categories). Defines the LaTeX environments (`faqpair`, `featureitem`) and cross-referencing system (`\faqref{}`, `\featureref{}`).

6.2 Framework Guides

`four-risks.md`
: Cagan's four risks (value, usability, feasibility, viability) mapped to PR/FAQ sections. Includes low/high risk signals, seven review criteria, and eight named decision outcomes (Go, Not Differentiated, TAM Too Small, Investment Too Risky, Not Feasible, Not Viable, Deprioritized, Iterate).

`common-mistakes.md`
: Seven anti-patterns: skills-forward thinking, confusing speed with velocity, selling instead

of truth-seeking, discounting competition, vague customer definition, great product wrong problem, and false precision in timeline estimates. Each has detection signals and fixes.

`decision-quality.md`
Twelve cognitive biases from Kahneman, Lovallo, and Sibony (HBR 2011) adapted to PR/FAQ review: self-interest, affect heuristic, groupthink, saliency, confirmation, availability, anchoring, halo effect, sunk cost, overconfidence, disaster neglect, and loss aversion. Each includes PR/FAQ-specific detection signals.

6.3 Meeting Guide

`meeting-guide.md`
Defines the four personas, the meeting flow (pre-meeting scan → agenda → per-hot-spot debate → summary), synthesis guidelines for turning parallel agent outputs into narrative, and hive mode consensus rules. Hot spots are classified as one-way doors (irreversible) or two-way doors (reversible), with door-weighted voting: one-way doors give Wei and Alex veto power; two-way door ties resolve in favor of action.

6.4 Risk Lens Guides

Three guides provide deep expertise for specific risk dimensions, each used primarily by one meeting persona:

`principal-engineer.md` (Wei)
Feasibility risk. Covers architecture trade-offs (monolith vs. services, build vs. buy, consistency vs. availability), operational complexity at scale, data model decisions as the most expensive to reverse, and dependency risk taxonomy. Core question: "What is the hardest unsolved technical problem, and does the PR/FAQ acknowledge it?"

`unit-economics.md` (Alex)
Viability risk. Defines CAC, LTV, LTV:CAC ratio, payback period, gross margin, and contribution margin. Distinguishes when unit economics are primary (SaaS, marketplace) from when they are secondary (internal tools, open source, strategic moats). Includes concrete benchmarks (LTV:CAC above 3:1, payback under 12 months, gross margin above 60%).

`ux-bar-raiser.md` (Priya)
Usability risk. Evaluates the full customer journey in five stages: awareness, evaluation, onboarding, first value, habit formation. Covers cognitive load sources, mental model alignment, the three-step Getting Started test, accessibility minimums, and error recovery principles.

6.5 Writing Quality Guide

`precise-writing.md`
Editorial standard for the streamliner agent. Adapted from Vervago's Precision Q+A and Amazon's "Write Like an Amazonian." Defines what to cut (redundancy, weasel words, hollow adjectives, hedge stacking, inflated phrases, throat-clearing sentences) and what to preserve (citations, customer quotes, risk assessments, numbers, structural elements). Target: 10–20% shorter with zero weasel words.

7. Stage Calibration

Every document declares its stage via `\prfaqstage{value}`:

hypothesis Pre-validation. Soft evidence acceptable. Focus on whether the author has identified the riskiest assumption and proposed a test.

validated Post-interviews or post-testing. Quantitative evidence and specific metrics expected. Primary customer data required in FAQs.

growth Post-launch with real users. All claims data-backed. Usage analytics, churn rates, and unit economics from actual operations.

Every reference guide, every agent, and every meeting persona adjusts its behavior based on the document's stage. The stage calibration tables at the end of each guide define these adjustments. This means the same guide produces three levels of rigor without branching logic in the agent prompts.

8. Build System

The quality gate is compilation of every LaTeX file in the repo's compile directories (`.`, `plugin/assets`, `docs`), which the `Makefile` globs rather than lists, so a new document is gated the moment it exists. Today that is five files:

1. `prfaq.tex` — the plugin's own PR/FAQ (dogfood document)
2. `press-release-v1.0.0.tex` — the externalized v1.0.0 release
3. `plugin/assets/prfaq-template.tex` — the template used to generate new documents
4. `plugin/assets/press-release-template.tex` — the template `/prfaq:externalize` builds from
5. `docs/prfaq-overview.tex` — this document

All must compile to valid PDFs before every commit. `make prfaq` runs the list; `compile_prfaq.sh` compiles one file (`pdflatex` → `biber` → `pdflatex` × 2). The `.docx` export pipeline preprocesses custom LaTeX environments into standard LaTeX, then hands off to pandoc.

9. Key Design Decisions

1. **Agents as domain experts.** Each of the eight agents has a narrow role with specific reference guides. The meeting agents embody named personas with competing risk lenses from Cagan's framework.
2. **Reference guides as encoded knowledge.** Domain expertise is defined once in standalone guides and loaded by multiple agents. No duplication between skill prompts and agent prompts.
3. **Stage-aware everything.** A single `\prfaqstage{}` declaration calibrates every agent, reviewer, and meeting persona. Hypothesis documents get softer scrutiny than growth documents.
4. **LaTeX as the artifact format.** Custom environments (`faqpair`, `featureitem`), cross-references (`\faqref{}`, `\featureref{}`), and bibliography support via biblatex/biber produce a professional, cross-referenced document.

5. **No runtime code.** The plugin is pure prompt engineering. Shell scripts handle compilation and export. Claude Code provides all execution capability.